

CS2311 Modern Operating System Exercises Spring 2026

Exercise 3: Deadlocks

Due date: April 21, 2026

- Suppose there are 2 copies of resource A, 3 copies of resource B, and 3 copies of resource C. Suppose further that thread 1 holds one unit of resources B and C and is waiting for a unit of A; that thread 2 is holding a unit of A and waiting on a unit of B; and that thread 3 is holding one unit of A, two units of B, and one unit of C. Draw the resource allocation graph. Is the system in a deadlocked state? Why or why not?
- Consider the following system snapshot using data structures in the Banker's algorithm, with resources A, B, C, and D, and thread P0 to P4:

	Max				Allocation				Need				Available			
	A	B	C	D	A	B	C	D	A	B	C	D	A	B	C	D
P0	6	0	1	2	4	0	0	1								
P1	1	7	5	0	1	1	0	0								
P2	2	3	5	6	1	2	5	4								
P3	1	6	5	3	0	6	3	3								
P4	1	6	5	6	0	2	1	2								
													3	2	1	1

Using Banker's algorithm, answer the following questions.

- How many resources of type A, B, C, and D are there?
 - What are the contents of the Need matrix?
 - Is the system in a safe state? Why?
 - If a request from thread P4 arrives for additional resources of (1,2,0,0), can the Banker's algorithm grant the request immediately? Show the new system state and other criteria.
- Consider a system with four threads T1, T2, T3, and T4, and two resources, R1, and R2, respectively. Each resource has two instances. Furthermore:
 - T1 allocates an instance of R2, and requests an instance of R1.
 - T2 allocates an instance of R1, and doesn't need any other resource.
 - T3 allocates an instance of R1 and requires an instance of R2.
 - T4 allocates an instance of R2, and doesn't need any other resource
 - Draw the resource allocation graph.
 - Is there a cycle in the graph? If yes name it.
 - Is the system in deadlock? If yes, explain why. If not, give a possible sequence of executions after which every thread completes.
 - A system has four threads and five allocable resources. The current allocation and maximum needs are as follows:

	Allocated	Maximum	Available
Thread A	1 0 2 1 1	1 1 2 1 3	0 0 1 X 2
Thread B	2 0 1 1 0	2 2 2 1 0	

Thread C 1 1 0 1 0 2 1 3 1 0

Thread D 1 1 1 1 0 1 1 2 2 1

What is the smallest value of X for which this is a safe state?